

IIIT-Hyd PGEE Fastrack Batch

Algorithms

Lecture_ 01



By- Aditya Sir

Topics to be Covered



Topic

Sorting Algorithms

Topic





About Aditya Jain sir

(AJ Sir)



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored working professions in field of Data Science and Analytics
11. Have been mentoring GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.

Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : (Lecture Schedule)



1. Analysis of Algorithms

1. Asymptotic Notations
2. Analysing Non-Recursive Algorithms
3. Analysing Loops
4. Analysing Recursive Algorithms
5. Space Complexity
6. Problem Solving



Topic : (Lecture Schedule)

[Course Planner]



- 1) TC analysis $\rightarrow$ Part 1
- 2) TC analysis $\rightarrow$ Part 2
- 3) Sorting Algorithms
- 4) Divide & Conquer - 1
- 5) Divide & Conquer - 2
- 6) Greedy method - 1
- 7) Greedy method - 2
- 8) DP
- 9) Graphs



Topic : Discrete Properties of ASN



ASN
Properties

Reflexive

	O	Ω	θ	o	ω
Reflexive	✓	✓	✓	✗	✗
Symmetric	✗	✗	✓	✗	✗
Transitive	✓	✓	✓	✓	✓
Transpose	if $f(n)$ is	$O(g(n))$	✗	if $f(n)$ is	$O(g(n))$
Symmetry	Then $g(n)$	is $\Omega(f(n))$	✗	Then $g(n)$	is $\omega(f(n))$

$a <'$

$c \rightarrow a < c$

Topic : Exponentials

For all real $a > 0$, m , n

$$a^0 = 1$$

$$a^1 = a$$

$$a^{-1} = 1/a$$

$$(a^m)^n = a^{mn}$$

$$(a^m)^n = (a^n)^m$$

$$a^m \cdot a^n = a^{m+n}$$

$$\left[(a^m)^n = (a^n)^m = a^{n \times m} \right]$$

Topic : Analysis of Algorithms

$$\log X^y = y \log x$$

$$\log xy = \log x + \log y$$

$$\log \log n = \log (\log n)$$

$$a^{\log_b x} = x^{\log_b a}$$

$$a = b^{\log_b a} = a^{\log_b b} = a$$

$$\log_b a^n = n \cdot \log_b a$$

$$\log_b a = \frac{1}{\log_a b}$$

$$\log n = \log_{10} n$$

$$\log^k n = \log n \rightarrow (\log n)^k$$

$$\log \frac{x}{y} = \log x - \log y$$

$$\log_b^x = \frac{\log_a^x}{\log_a^b}$$

$$\log_c(ab) = \log_c a + \log_c b$$

$$\log_b^{1/a} = -\log_b a$$

$$\log_b a = \log_a b$$

$$\log_b(a^n) = n \log_b a$$

$$\begin{aligned}\log_b\left(\frac{1}{a}\right) &= \log_b 1 - \log_b a \\ &= 0 - \log_b a \\ &= -\log_b a\end{aligned}$$

Question



#Q. Consider the following functions:

$$f_1 = 2^{2^n}$$

$$f_2 = n!$$

$$f_3 = 4^n$$

$$f_4 = 2^n$$

$$2^{2^n} > 2^n$$
$$f_1 > f_4$$

What is the correct increasing order of above functions?

D

A $f_1 f_4 f_3 f_2$ ~~X~~

B $f_4 f_2 f_3 f_1$ ~~X~~

C $f_1 f_2 f_3 f_4$ ~~X~~

D $f_4 f_3 f_2 f_1$ ✓

Question



#Q. Consider the following functions from positive integers to real number:

$$10, \sqrt{n}, n, \log_2 n, \frac{100}{n}$$

The correct arrangement of the above functions in increasing order of asymptotic complexity is:

©

A $\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$ ✗

B $10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$ ✗

C $\frac{100}{n}, 10, \log_2 n, \sqrt{n}, n$ ✓

D $\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$ ✗

$$\begin{array}{c}
 10 \\
 \sqrt{n} \\
 n \\
 \log n \\
 \frac{100}{n}
 \end{array}$$



$$\frac{100}{n} < 10 < \log n < \sqrt{n} < n$$

~~~~~

## Question



#Q. Consider the following code:

```
main()
{
    i = 1;
    while (i ≤ n)
    {
        i = i * 2;
    }
}
```

C

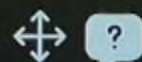
What is the time complexity of above code?

**A**  $\theta(n)$  ~~X~~

**B**  $\theta\sqrt{n}$  ~~X~~

**C**  $\theta(\log n)$

**D** None ~~X~~





## Question



#Q. Consider the following code:

```
main()
{
    i = 1;
    while (i ≤ n)
    {
        i = i * 2;
    }
}
```

C

75%

What is the time complexity of above code?

A  $\theta(n)$  X

B  $\theta\sqrt{n}$  X

C  $\theta(\log n)$

D None X

## Question

#Q. Consider the following code

```
i = n;  
while (i > 0)  $\longrightarrow \log n$   
{  
    j = 1;  
    while (j ≤ n)  $\longrightarrow \log n$   
    {  
        j = 2 * j;  
    }  
    i = i/2;  
}
```

**A**

$(\log n)^2$

**B**

$\sqrt{\log n}$

**C**

$n \log n$

**D**

$\log \log n$

$$\underline{\underline{TC = (\log n)^2}}$$

Time complexity of above code in terms of Big-Oh?



## Question

#Q.  $f(n) = 2^n$ ;  $g(n) = n^n$

$$2^n \quad n^n$$

$$2 < n$$

$$2^n < n^n$$

**A**  $f(n) = O(g(n))$

$$f \leq g$$

**B**  $f(n) = \theta(g(n))$

$$f = g$$

**C**  $f(n) = \Omega(g(n))$

$$f \geq g$$

**D** None of these

## Question



Imp

#Q.  $f(n) = n \cdot 2^n$ ;  $g(n) = 4^n$

**A**  $f(n) = O(g(n))$

**B**  $f(n) = \theta(g(n))$

**C**  $f(n) = \Omega(g(n))$

**D** None of these



## Question



#Q.  $f(n) = n \cdot 2^n$ ;  $g(n) = 4^n$

Imp



**A**  $f(n) = O(g(n))$

$f \leq g$

A

**B**  $f(n) = \theta(g(n))$

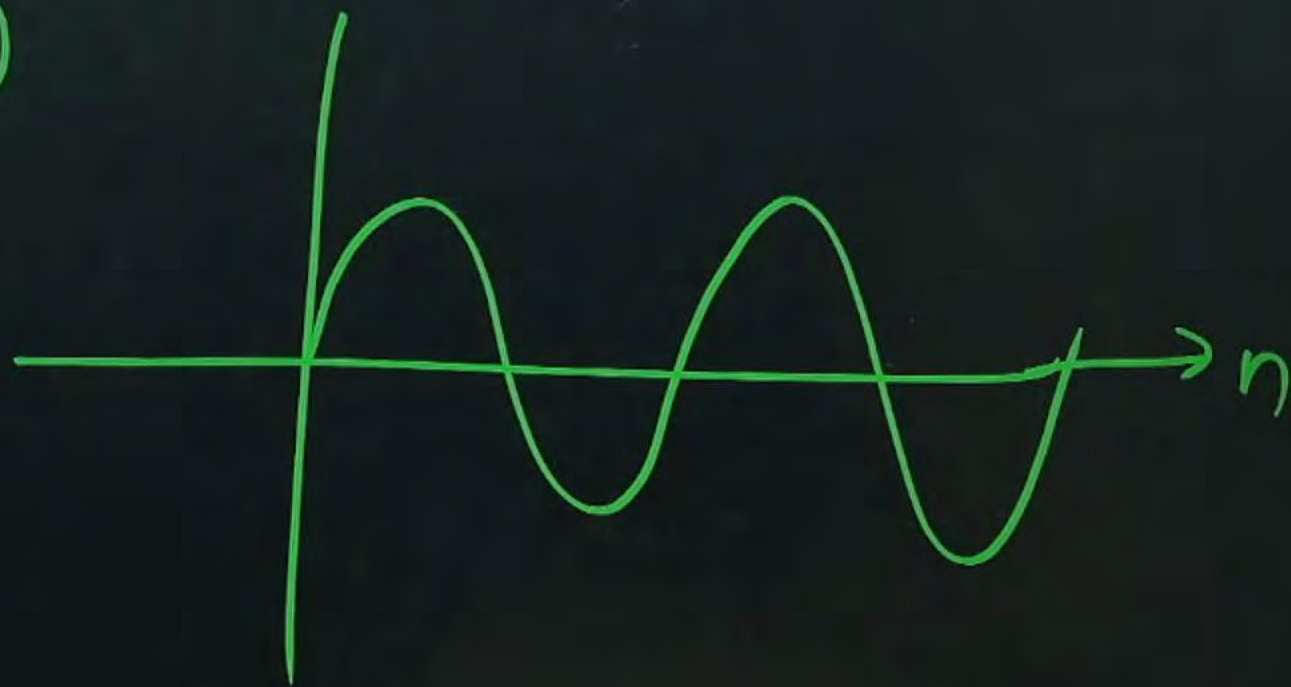
$f = g$

**C**  $f(n) = \Omega(g(n))$

$f \geq g$

**D** None of these

$$f = n$$
$$g = n^{(1 + \sin(n))}$$





C1

$$\sin n = -1$$

$$\# = n \quad g = n^{(1+(-1))}$$

$$\boxed{f = \Omega(g)} = \underline{\underline{1}}$$

C2:

C1

$$\sin n = -1$$

$$f = n \quad g = n^{(1+(-1))}$$

$$\boxed{f = \Omega(g)} = \underline{\underline{1}} = n^0$$

C2:

$$\sin n = 1$$

$$f = n, \quad g = n^{(1+1)}$$

$$= n^2$$

$$f < g \Rightarrow \underline{\underline{f = O(g)}}$$







## Topic : (Running Times of Program Segments with Loops Program)

- |                                     | Times run  | T.C             |
|-------------------------------------|------------|-----------------|
| 1. for (i = 1; i <= n; i = i + 1) → | $n$        | $O(n)$          |
| 2. for (i = 1; i <= n; i = i + 2) → | $n/2$      | $O(n/2) = O(n)$ |
| 3. for (i = 1; i <= n; i = i + 5) → | $n/5$      | $O(n/5) =$      |
| 4. for (i = 1; i <= n; i = i * 2) → | $\log_2 n$ | $O(\log)$       |
| 5. for (i = 1; i <= n; i = i * 3) → | $\log_3 n$ | $O(\log)$       |

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## Question



#Q. Consider the following function:

```
int unknown (int n)
```

```
{
```

```
    int i, j, k = 0;
```

```
    for (i=n/2; i <= n; i++)
```

```
        for(j = 2; j <= n; j = j*2)
```

```
            k = k + n/2;
```

```
    return(k);
```

```
}
```

The return value of the function is

**A**

$\theta(n^2)$

**B**

$\theta(n^3)$

**C**

$\theta(n^2 \log n)$

**D**

$\theta(n^3 \log n)$



?

Adv

$n/2 \rightarrow n \quad (+1) \rightarrow O(n)$

$2 \rightarrow n \quad (*2) \rightarrow O(\log_2 n)$

loop:  $O(n \log n)$

$$K = K + n/2$$

$$\Rightarrow \left[ n \log n \times n/2 \right] \\ = \left( \frac{n^2 \log n}{2} \right)$$





# Topic : Asymptotic Notations



## Solving recurrence

1. Back substitution

2. Master Method

3. Recursion tree

Shortcut

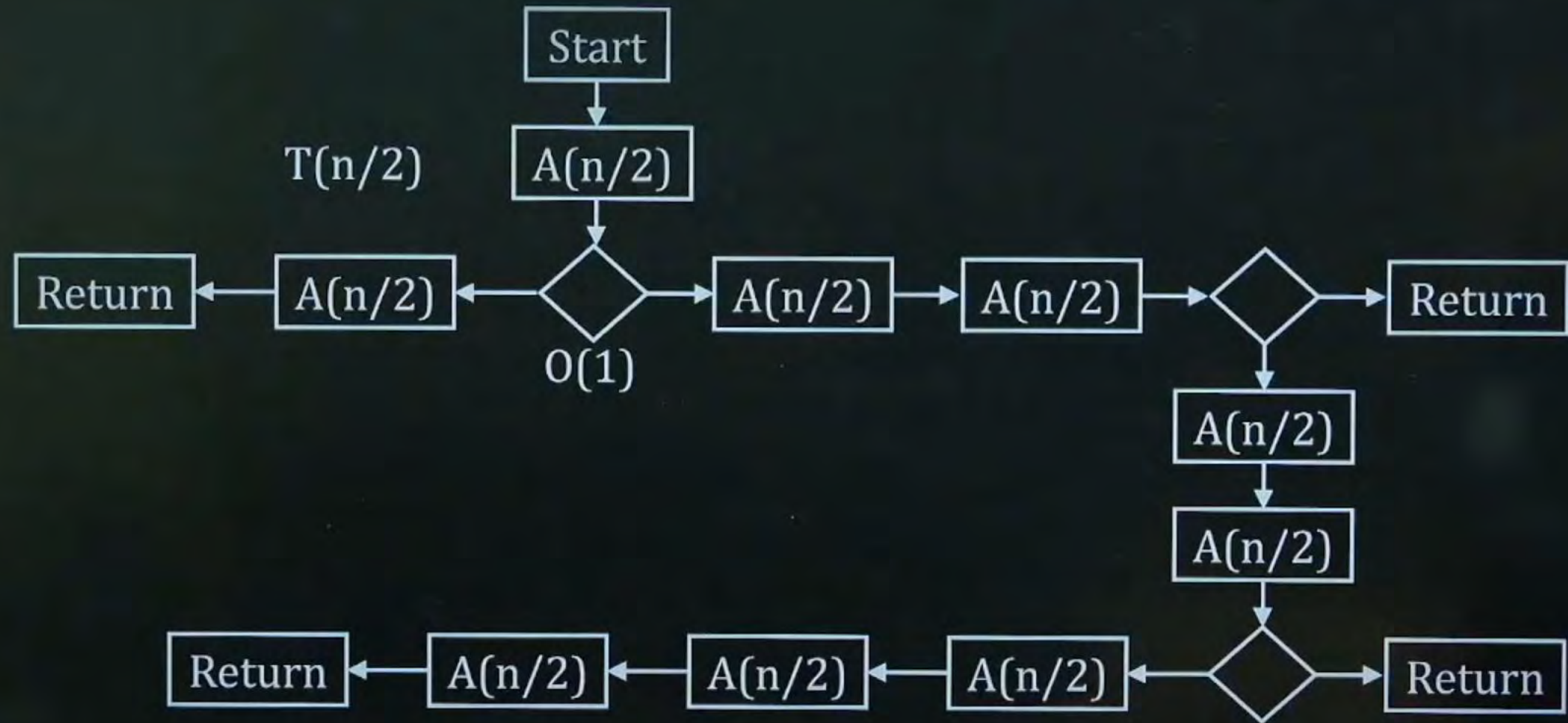
1) value  
2)  $TC = O()$

$TC = O()$

$TC = O()$



## Topic : Time Complexity Framework for Recursive Algorithms







**THANK - YOU**